

Math Logic File: Model, Lemmas, and Proofs

Multiple Nuisance Breaks and Spurious Predictability under Persistent Regressors

1 Model Setup and Score Objects

Let

$$\mathbf{Y} = (Y_1, \dots, Y_T)', \quad \mathbf{X}_{lag} = (X_0, \dots, X_{T-1})',$$

and let $\mathbf{w} = (w_1, \dots, w_T)'$, where $w_t = w(X_{t-1})$ is a bounded saturated weight. The predictor satisfies

$$X_t = \theta + \rho_T X_{t-1} + \varepsilon_t. \quad (1.1)$$

The maintained outcome model is

$$Y_t = \beta_0 X_{t-1} + \sum_{j=0}^{q_0} m_j(W_{t-1}) \mathbf{1}\{k_{j,0} < t \leq k_{j+1,0}\} + U_t, \quad (1.2)$$

where

$$0 = k_{0,0} < k_{1,0} < \dots < k_{q_0,0} < k_{q_0+1,0} = T.$$

The coefficient β_0 is stable over the full sample. The structural breaks are confined to the nonparametric nuisance component.

For a candidate partition $\Lambda = (k_1, \dots, k_q)$, write $k_0 = 0$, $k_{q+1} = T$, and

$$I_j(\Lambda) = \{t : k_j < t \leq k_{j+1}\}.$$

The true partition is $\Lambda_0 = (k_{1,0}, \dots, k_{q_0,0})$. Define $j_\Lambda(t) = j$ when $t \in I_j(\Lambda)$. The true nuisance vector is

$$\mu_t = m_{j_{\Lambda_0}(t)}(W_{t-1}), \quad \boldsymbol{\mu} = (\mu_1, \dots, \mu_T)'.$$

Let $\mathbf{P}$ be the $T \times K$ sieve matrix whose t -th row is

$$\mathbf{P}_K(W_{t-1})' = (p_1(W_{t-1}), \dots, p_K(W_{t-1})).$$

For a partition Λ , let $\mathbf{D}_j(\Lambda)$ be the diagonal matrix with diagonal entry $\mathbf{1}\{t \in I_j(\Lambda)\}$. Define

$$\mathbf{Q}_{K,\Lambda} = [\mathbf{D}_0(\Lambda)\mathbf{P}, \mathbf{D}_1(\Lambda)\mathbf{P}, \dots, \mathbf{D}_q(\Lambda)\mathbf{P}], \quad (1.3)$$

and

$$\mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{Q}_{K,\Lambda}(\mathbf{Q}_{K,\Lambda}'\mathbf{Q}_{K,\Lambda})^{-1}\mathbf{Q}_{K,\Lambda}'. \quad (1.4)$$

The block nuisance space is

$$\mathcal{N}_{K,\Lambda} = \left\{ \sum_{j=0}^q \mathbf{1}(t \in I_j(\Lambda)) \mathbf{P}_K(W_{t-1})' \mathbf{c}_j : \mathbf{c}_j \in \mathbb{R}^K \right\}.$$

The matrix $\mathbf{M}_{K,\Lambda}$ is the orthogonal residual-maker for this empirical space.

For each β and Λ , define the profiled residual, regime-orthogonalized weight, and scalar score by

$$\hat{\mathbf{u}}(\beta, \Lambda) = \mathbf{M}_{K, \Lambda}(\mathbf{Y} - \beta \mathbf{X}_{lag}), \quad (1.5)$$

$$\mathbf{w}^c(\Lambda) = \mathbf{M}_{K, \Lambda} \mathbf{w}, \quad (1.6)$$

$$\hat{Z}_t(\beta, \Lambda) = \hat{u}_t(\beta, \Lambda) w_t^c(\Lambda). \quad (1.7)$$

The empirical likelihood ratio is

$$L_T(\beta, \Lambda) = \sup \left\{ \prod_{t=1}^T (T\pi_t) : \pi_t \geq 0, \sum_{t=1}^T \pi_t = 1, \sum_{t=1}^T \pi_t \hat{Z}_t(\beta, \Lambda) = 0 \right\}, \quad (1.8)$$

$$\ell_T(\beta, \Lambda) = -2 \log L_T(\beta, \Lambda) = 2 \sum_{t=1}^T \log \{1 + \hat{\lambda}(\beta, \Lambda) \hat{Z}_t(\beta, \Lambda)\}, \quad (1.9)$$

where $\hat{\lambda}(\beta, \Lambda)$ solves the scalar multiplier equation.

For an interval $[s, e]$, define the null-imposed segment cost

$$\mathcal{C}_K(s, e; \beta) = \min_{\mathbf{c} \in \mathbb{R}^K} \sum_{t=s}^e \{Y_t - \beta X_{t-1} - \mathbf{P}_K(W_{t-1})' \mathbf{c}\}^2. \quad (1.10)$$

For $\Lambda = (k_1, \dots, k_q)$, define

$$RSS_K(\beta, \Lambda) = \sum_{j=0}^q \mathcal{C}_K(k_j + 1, k_{j+1}; \beta). \quad (1.11)$$

Let $\mathcal{L}_q(\epsilon)$ be the set of q -break partitions satisfying minimum spacing. The profile partition estimator is

$$\hat{\Lambda}_q(\beta) = \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} RSS_K(\beta, \Lambda). \quad (1.12)$$

If the number of breaks is unknown, define

$$\hat{q}(\beta) = \arg \min_{0 \leq q \leq \bar{q}} \left\{ RSS_K(\beta, \hat{\Lambda}_q(\beta)) + \text{pen}_T(q, K) \right\}, \quad (1.13)$$

and set

$$\hat{\Lambda}(\beta) = \hat{\Lambda}_{\hat{q}(\beta)}(\beta).$$

The break-adaptive statistic is

$$\ell_T^{BA}(\beta) = \ell_T\{\beta, \hat{\Lambda}(\beta)\}. \quad (1.14)$$

2 Assumptions

Assumption 1 (Finite nuisance breaks and smooth regimes). *The number q_0 is fixed. For some $\epsilon > 0$,*

$$\min_{0 \leq j \leq q_0} (k_{j+1,0} - k_{j,0}) \geq \epsilon T.$$

Each m_j belongs to the same smoothness class $\mathcal{H}^s$. If $q_0 \geq 1$, the adjacent breaks satisfy

$$\Delta_T = \min_{0 \leq j < q_0} \|m_{j+1} - m_j\|_{L^2(P_W)} > 0,$$

with $T\Delta_T^2$ large enough relative to the block sieve dimension and logarithmic search complexity. A sufficient fixed-break version is $T\Delta_T^2/(K \log T) \rightarrow \infty$. When $q_0 = 0$, this condition is void.

Assumption 2 (Identification after block residualization). *For the true partition,*

$$T^{-1} \mathbf{w}' \mathbf{M}_{K, \Lambda_0} \mathbf{w} \xrightarrow{p} Q_w, \quad P(Q_w > \underline{q}) = 1$$

for some $\underline{q} > 0$.

Assumption 3 (Block-sieve rates). *For the true partition and each regime j , there exists $\mathbf{c}_{j,K}$ such that*

$$m_j(W_{t-1}) = \mathbf{P}_K(W_{t-1})' \mathbf{c}_{j,K} + r_{j,K,t},$$

with $\max_j \|\mathbf{r}_{j,K}\|_T = O_p(a_K)$, $\max_j \|\mathbf{r}_{j,K}\|_\infty = O_p(a_K)$, and $\sqrt{T}a_K \rightarrow 0$. The block Gram matrix $T^{-1} \mathbf{Q}'_{K, \Lambda_0} \mathbf{Q}_{K, \Lambda_0}$ has eigenvalues bounded away from zero and infinity with probability tending to one. If $\zeta_K = \max_t \|\mathbf{P}_K(W_{t-1})\|$, then

$$\frac{(q_0 + 1)K}{\sqrt{T}} \rightarrow 0, \quad \zeta_K \sqrt{\frac{(q_0 + 1)K}{T}} \rightarrow 0.$$

Assumption 4 (Known-partition oracle score). *With $\mathbf{v}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{w}$, define $\tilde{\mathbf{U}}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{U}$ and*

$$A_T^{(q)} = T^{-1/2} \sum_{t=1}^T U_t v_{t,0}, \quad V_T^{(q)} = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2.$$

The oracle numerator and denominator satisfy

$$A_T^{(q)} \xrightarrow{\text{st}} MN(0, \eta_q^2), \quad V_T^{(q)} \xrightarrow{p} \eta_q^2, \quad \max_{t \leq T} |U_t v_{t,0}| = o_p(\sqrt{T}),$$

where $P(\eta_q^2 > \underline{\eta}) = 1$ for some $\underline{\eta} > 0$. In addition,

$$T^{-1} \sum_{t=1}^T \{(\tilde{U}_{0,t} v_{t,0})^2 - (U_t v_{t,0})^2\} = o_p(1), \quad \max_{t \leq T} |(\tilde{U}_{0,t} - U_t) v_{t,0}| = o_p(\sqrt{T}).$$

The scalar empirical-likelihood convex-hull condition holds under the null.

Assumption 5 (Partition estimation). *For the true number of breaks,*

$$\max_{1 \leq j \leq q_0} |\hat{k}_j(\beta_0) - k_{j,0}| = O_p(r_T), \quad r_T = o(\sqrt{T}).$$

Equivalently, the number of observations assigned to the wrong regime satisfies

$$|\mathcal{M}_T| = \left| \{t : j_{\hat{\Lambda}(\beta_0)}(t) \neq j_{\Lambda_0}(t)\} \right| = o_p(\sqrt{T}).$$

Let

$$\mathcal{Z}_T = \max_{t \leq T} \{|\hat{Z}_t(\beta_0, \hat{\Lambda}(\beta_0))| + |\hat{Z}_t(\beta_0, \Lambda_0)|\}.$$

The boundary observations are negligible at the numerator and denominator scales:

$$T^{-1/2} |\mathcal{M}_T| \mathcal{Z}_T = o_p(1), \quad T^{-1} |\mathcal{M}_T| \mathcal{Z}_T^2 = o_p(1).$$

3 Statements and Proofs

The stable no-break score is

$$S_T^{\text{st}}(\beta_0) = T^{-1/2}(\mathbf{Y} - \beta_0 \mathbf{X}_{\text{lag}})' \mathbf{M}_{K,0} \mathbf{w},$$

where 0 denotes the no-break partition.

Proposition 1 (Stable-score decomposition). *Under (1.2), the stable-nuisance score satisfies the exact identity*

$$S_T^{\text{st}}(\beta_0) = A_T^{\text{st}} + B_T,$$

where

$$A_T^{\text{st}} = T^{-1/2} \mathbf{U}' \mathbf{M}_{K,0} \mathbf{w}, \quad B_T = T^{-1/2} \boldsymbol{\mu}' \mathbf{M}_{K,0} \mathbf{w}.$$

The term B_T is the omitted-break component left by imposing the no-break nuisance projection.

Proof. Under $H_0 : \beta = \beta_0$,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{\text{lag}} = \boldsymbol{\mu} + \mathbf{U}.$$

Multiplying by the stable residualized weight gives

$$T^{-1/2}(\mathbf{Y} - \beta_0 \mathbf{X}_{\text{lag}})' \mathbf{M}_{K,0} \mathbf{w} = T^{-1/2} \mathbf{U}' \mathbf{M}_{K,0} \mathbf{w} + T^{-1/2} \boldsymbol{\mu}' \mathbf{M}_{K,0} \mathbf{w}.$$

This is an exact algebraic identity. No stable-sieve approximation remainder is needed because B_T is defined using the exact nuisance vector $\boldsymbol{\mu}$. $\square$

Definition 1 (Break-induced spurious predictivity). The stable nuisance specification generates break-induced spurious predictivity at β_0 if

$$T^{-1/2} \langle \mathbf{M}_{K,0} \boldsymbol{\mu}, \mathbf{M}_{K,0} \mathbf{w} \rangle_{\mathbb{R}^T} = B_T \neq o_p(1).$$

The condition is directional: a nuisance break with little projection on the residualized persistent weight need not distort the test.

Theorem 1 (Omitted-break failure). *Suppose the stable empirical-likelihood statistic admits the scalar expansion*

$$\ell_T^{\text{st}}(\beta_0) = \frac{\{S_T^{\text{st}}(\beta_0)\}^2}{V_T^{\text{st}}} + o_p(1),$$

with V_T^{st} bounded away from zero in probability. If

$$|B_T| / (V_T^{\text{st}})^{1/2} \xrightarrow{p} \infty,$$

then, for every fixed chi-square critical value c_α ,

$$P\{\ell_T^{\text{st}}(\beta_0) > c_\alpha\} \rightarrow 1.$$

If instead

$$A_T^{\text{st}} / (V_T^{\text{st}})^{1/2} \Rightarrow N(0, 1), \quad B_T / (V_T^{\text{st}})^{1/2} \Rightarrow B, \quad P(B \neq 0) > 0,$$

then the stable statistic has a noncentral, or random-noncentral, chi-square limit rather than a central χ_1^2 limit.

Proof. By Proposition 1,

$$S_T^{\text{st}}(\beta_0) = A_T^{\text{st}} + B_T.$$

If $|B_T|/(V_T^{\text{st}})^{1/2} \rightarrow_p \infty$, the quadratic ratio in the scalar expansion diverges in probability, so rejection at any fixed critical value has probability tending to one. If the normalized oracle term converges to $N(0, 1)$ and the normalized break term converges to a nondegenerate random variable B , the limiting quadratic form is $(Z + B)^2$, where Z is standard normal in the limiting self-normalized score. This is central chi-square only when $B = 0$ with probability one. $\square$

Lemma 1 (Block projection identities). *If $\mathbf{Q}'_{K,\Lambda} \mathbf{Q}_{K,\Lambda}$ is nonsingular, then*

$$\mathbf{Q}'_{K,\Lambda} \mathbf{M}_{K,\Lambda} = \mathbf{0}, \quad \mathbf{M}'_{K,\Lambda} = \mathbf{M}_{K,\Lambda}, \quad \mathbf{M}_{K,\Lambda}^2 = \mathbf{M}_{K,\Lambda}.$$

For every $\mathbf{c}$ conformable with $\mathbf{Q}_{K,\Lambda}$,

$$(\mathbf{Q}_{K,\Lambda} \mathbf{c})' \mathbf{M}_{K,\Lambda} \mathbf{w} = 0.$$

Proof. Let

$$\mathbf{\Pi}_{K,\Lambda} = \mathbf{Q}_{K,\Lambda} (\mathbf{Q}'_{K,\Lambda} \mathbf{Q}_{K,\Lambda})^{-1} \mathbf{Q}'_{K,\Lambda}.$$

This is the orthogonal projector onto the column space of $\mathbf{Q}_{K,\Lambda}$. Hence $\mathbf{\Pi}'_{K,\Lambda} = \mathbf{\Pi}_{K,\Lambda}$, $\mathbf{\Pi}_{K,\Lambda}^2 = \mathbf{\Pi}_{K,\Lambda}$, and $\mathbf{Q}'_{K,\Lambda} (\mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda}) = \mathbf{0}$. Since $\mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda}$, the displayed identities follow. The final equality is obtained by multiplying $\mathbf{Q}'_{K,\Lambda} \mathbf{M}_{K,\Lambda} = \mathbf{0}$ by $\mathbf{c}$ and $\mathbf{w}$. $\square$

Set $\Lambda = \Lambda_0$ and write

$$\mathbf{v}_0 = \mathbf{M}_{K,\Lambda_0} \mathbf{w}, \quad \hat{\mathbf{u}}_0(\beta_0) = \mathbf{M}_{K,\Lambda_0} (\mathbf{Y} - \beta_0 \mathbf{X}_{lag}).$$

Under the true partition,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \mathbf{Q}_{K,\Lambda_0} \mathbf{c}_0 + \mathbf{r}_0 + \mathbf{U},$$

where $\mathbf{r}_0$ stacks the regime-specific sieve approximation errors.

Lemma 2 (Known-partition replacement). *Under Assumptions 3 and 4,*

$$T^{-1/2} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^T U_t v_{t,0} + o_p(1), \quad (3.1)$$

$$T^{-1} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda_0)^2 = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2 + o_p(1), \quad (3.2)$$

$$\max_{t \leq T} |\hat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}). \quad (3.3)$$

Proof. Using Lemma 1,

$$\hat{\mathbf{u}}_0(\beta_0) = \mathbf{M}_{K,\Lambda_0} (\mathbf{U} + \mathbf{r}_0), \quad \mathbf{v}_0 = \mathbf{M}_{K,\Lambda_0} \mathbf{w}.$$

Because $\mathbf{M}_{K,\Lambda_0}$ is symmetric and idempotent,

$$T^{-1/2}\widehat{\mathbf{u}}_0(\beta_0)'\mathbf{v}_0 = T^{-1/2}(\mathbf{U} + \mathbf{r}_0)'\mathbf{v}_0.$$

The approximation term obeys

$$|T^{-1/2}\mathbf{r}_0'\mathbf{v}_0| \leq \sqrt{T}\|\mathbf{r}_0\|_T\|\mathbf{v}_0\|_T = o_p(1),$$

which proves (3.1). For the denominator, write $\widetilde{\mathbf{U}}_0 = \mathbf{M}_{K,\Lambda_0}\mathbf{U}$. The feasible score decomposes as

$$\widehat{Z}_t(\beta_0, \Lambda_0) = \widetilde{U}_{0,t}v_{t,0} + (\mathbf{M}_{K,\Lambda_0}\mathbf{r}_0)_tv_{t,0}.$$

The approximation contribution is negligible by Assumption 3. Assumption 4 then gives

$$T^{-1}\sum_{t=1}^T\{(\widetilde{U}_{0,t}v_{t,0})^2 - (U_tv_{t,0})^2\} = o_p(1),$$

which proves (3.2). The maximum bound follows from the oracle maximum condition, the projected-noise replacement condition, and the sup-norm sieve approximation bound. $\square$

Lemma 3 (Scalar empirical-likelihood expansion). *Let $Z_{t,T}$ be scalar scores. Suppose the convex-hull condition holds and*

$$T^{-1/2}\sum_t Z_{t,T} = O_p(1), \quad T^{-1}\sum_t Z_{t,T}^2 \rightarrow_p \eta^2, \quad \max_t |Z_{t,T}| = o_p(\sqrt{T}),$$

where $P(\eta^2 > \underline{\eta}) > 1 - o(1)$. Then

$$2\sum_{t=1}^T \log(1 + \widehat{\lambda}Z_{t,T}) = \frac{\{T^{-1/2}\sum_t Z_{t,T}\}^2}{T^{-1}\sum_t Z_{t,T}^2} + o_p(1).$$

Proof. The multiplier equation implies $\widehat{\lambda} = O_p(T^{-1/2})$. Since $\max_t |\widehat{\lambda}Z_{t,T}| = o_p(1)$, a Taylor expansion of the multiplier equation gives

$$\widehat{\lambda} = \frac{\sum_t Z_{t,T}}{\sum_t Z_{t,T}^2} + o_p(T^{-1/2}).$$

Substitution into the second-order expansion of the log empirical likelihood gives the displayed quadratic approximation. $\square$

Theorem 2 (Known-partition Wilks theorem). *Under Assumptions 1, 2, 3, and 4, under $H_0 : \beta = \beta_0$,*

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2.$$

Proof. Apply Lemma 2 to reduce the feasible score to the matched block-projection oracle score $U_tv_{t,0}$. Assumption 4 gives the mixed-normal numerator limit, denominator consistency, and maximum-score condition. Lemma 3 then yields

$$\ell_T(\beta_0, \Lambda_0) = \frac{\{T^{-1/2}\sum_t U_tv_{t,0}\}^2}{T^{-1}\sum_t U_t^2 v_{t,0}^2} + o_p(1) \Rightarrow \chi_1^2.$$

$\square$

Let $\hat{\Lambda} = \hat{\Lambda}(\beta_0)$ and

$$\mathcal{M}_T = \{t : j_{\hat{\Lambda}}(t) \neq j_{\Lambda_0}(t)\}.$$

Lemma 4 (Misclassification bound). *Suppose Assumption 5 holds. Suppose also that the projected weights are uniformly bounded in probability, the score arrays satisfy the same maximum-score bounds as in Assumption 4, and the segment Gram matrices are stable uniformly over admissible partitions. Then*

$$\begin{aligned} T^{-1/2} \sum_{t=1}^T \{\hat{Z}_t(\beta_0, \hat{\Lambda}) - \hat{Z}_t(\beta_0, \Lambda_0)\} &= o_p(1), \\ T^{-1} \sum_{t=1}^T \{\hat{Z}_t(\beta_0, \hat{\Lambda})^2 - \hat{Z}_t(\beta_0, \Lambda_0)^2\} &= o_p(1), \\ \max_t |\hat{Z}_t(\beta_0, \hat{\Lambda})| &= o_p(\sqrt{T}). \end{aligned}$$

Proof. Away from $\mathcal{M}_T$, the estimated partition and true partition assign observations to the same regime. The two residual makers differ only through least-squares fits perturbed by boundary observations and by terms controlled by uniform Gram stability. With $\mathcal{Z}_T$ defined in Assumption 5, the direct contribution of the boundary set to the numerator is bounded by

$$T^{-1/2} |\mathcal{M}_T| \mathcal{Z}_T = o_p(1).$$

The denominator contribution is bounded by $T^{-1} |\mathcal{M}_T| \mathcal{Z}_T^2 = o_p(1)$. The remaining terms are the uniform Gram-stability perturbations of the least-squares coefficients and are of the same smaller order. Uniform Gram stability transfers the maximum-score condition from the true partition to the estimated partition. $\square$

Theorem 3 (Estimated-partition equivalence). *Under the assumptions of Theorem 2 and Assumption 5,*

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} - \ell_T(\beta_0, \Lambda_0) = o_p(1).$$

Consequently,

$$\ell_T^{BA}(\beta_0) \Rightarrow \chi_1^2.$$

Proof. Lemma 4 gives matching numerator, denominator, and maximum-score behavior for the two score arrays.

Applying Lemma 3 to both arrays yields

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} - \ell_T(\beta_0, \Lambda_0) = o_p(1).$$

The Wilks limit follows from Theorem 2. $\square$

Corollary 1 (Unknown number of breaks). *If $P\{\hat{q}(\beta_0) = q_0\} \rightarrow 1$, then the conclusion of Theorem 3 remains valid with $\hat{\Lambda}(\beta_0)$ defined by (1.13).*

Proof. On the event $\{\hat{q}(\beta_0) = q_0\}$, the estimated-order partition is the fixed-order partition used in Theorem 3. Since this event has probability tending to one, the same first-order expansion and Wilks limit apply. $\square$

Theorem 4 (Local predictive alternatives). *Let $\beta_T = \beta_0 + d_T$, where $d_T = h/r_T^\beta$ is local to the information scale of the residualized persistent score. Suppose*

$$d_T T^{-1/2} \mathbf{X}'_{lag} \mathbf{M}_{K, \Lambda_0} \mathbf{w} \xrightarrow{p} \eta_q \delta_h,$$

with δ_h fixed, and suppose the known- and estimated-partition replacement conditions above continue to hold under β_T . Then, when the statistic is evaluated at β_0 ,

$$\ell_T^{BA}(\beta_0) \Rightarrow \chi_1^2(\delta_h^2).$$

The noncentrality parameter is generated by the predictive drift d_T , not by the nuisance-break drift B_T .

Proof. Under the local alternative,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \boldsymbol{\mu} + \mathbf{U} + d_T \mathbf{X}_{lag}.$$

At the true partition, the block projection removes the regime-specific sieve approximation to $\boldsymbol{\mu}$, so the score numerator has the expansion

$$T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^T U_t v_{t,0} + d_T T^{-1/2} \mathbf{X}'_{lag} \mathbf{M}_{K, \Lambda_0} \mathbf{w} + o_p(1).$$

By the local drift condition, the second term converges to $\eta_q \delta_h$. The denominator remains $\eta_q^2 + o_p(1)$ under the stated replacement conditions. Lemma 3 therefore gives a $\chi_1^2(\delta_h^2)$ limit for the known partition. Lemma 4 transfers the same numerator, denominator, and maximum-score behavior to $\widehat{\Lambda}(\beta_0)$, proving the estimated-partition version. $\square$